

Sub-processor Schedule — Compute and Signing Provider

This schedule governs processing carried out by the operator’s designated **Compute and Signing Provider** (the party fulfilling cloud infrastructure, application-layer signing, HSM period-close attestation, and account authentication). It is incorporated by reference into the Shyware Data Processing Agreement (`DATA_PROCESSING_AGREEMENT.tex`) and is published at <https://shyware.fyi/legal/privacy/dpia/dpa/schedule-compute>.

One provider may fulfill all three sub-roles or separate providers may be used per sub-role. The operator must name each and execute signed DPAs before processing personal data.

Configuration trigger: All deployments. Specific services depend on `signing.backend` and `api.auth_scheme` in the operator’s `shyconfig`.

Infrastructure Sub-role

Field	Value
Role	Cloud compute, networking, backups
Named provider	<code>REQUIRED_OPERATOR_INPUT</code>
Location	<code>REQUIRED_OPERATOR_INPUT</code>
Transfer mechanism	<code>REQUIRED_OPERATOR_INPUT</code>
Data processed	Server access logs, network metadata, backup snapshots

Signing Sub-role

Field	Value
Role	Application-layer signing (KMS) + HSM period-close attestation
Named provider	<code>REQUIRED_OPERATOR_INPUT</code>
<code>shyconfig</code> field	<code>signing.backend</code> : <code>aws_kms</code> , <code>aws_kms_x_aws_cloudhsm</code> , or equivalent
Data processed	Message digests for signing only — no payload content, no submission direction
Key material	Non-exportable; FIPS 140-3 L2 minimum (KMS); FIPS 140-3 L3 required for HSM period-close

Structural note: The signing provider holds no canonical ledger state and no off-chain linkage data. The HSM period-close signature over disjoint Merkle roots enables operator-independent verifiability — any third party can verify using only the HSM public key.

Account Authentication Sub-role

Field	Value
Role	Account authentication and session management
Named provider	REQUIRED_OPERATOR_INPUT
shyconfig field	api.auth_scheme: cognito, jwks, or equivalent
Data processed	Account credentials, session tokens, account sub claim

Privacy notice obligation: The account sub claim flows into participant identity hash derivation (List 2). The operator’s privacy policy must disclose that the authentication provider holds credentials that, combined with the reconciling authority’s data, could link accounts to canonical submissions. Assess in deployment DPIA.

Structural Constraints

- Signing provider processes digests only — no payload content, no identity-payload pairs
- HSM key is non-exportable; provider compromise does not expose historical attestations
- Authentication provider is not a data authority in the two-list protocol sense — it cannot write canonical state or access the off-chain linkage store unilaterally
- If all three sub-roles are fulfilled by one provider, document and assess the consolidation risk in the DPIA

Operator Obligations

- Execute signed DPAs with each named provider before processing
- Document providers, locations, and transfer mechanisms in the Records of Processing (`generate-rop.mjs`)
- Confirm SCCs or adequacy decision for any non-EEA provider
- Configure FIPS 140-3 L3 HSM backend before production
- Notify Shyware of provider changes at least 30 days in advance

This schedule is incorporated by reference into the Shyware Data Processing Agreement and is published at <https://shyware.fyi/legal/privacy/dpia/dpa/schedule-compute>. Swap by updating `signing.backend` and `api.auth_scheme` in `shyconfig.json` and executing new DPAs before transitioning.